



SOFTWARE DEVELOPMENT PLAN

Course Title : Software Project Design and Development

Course Code : CSE 416

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Introduction

This comprehensive report details the strategic plan for the **16-week software development project**. It summarizes the project's phased timeline, specific resource allocation model, and technology choices, confirming a successful budget constraint to **1,50,000 BDT** through strategic localization and efficient resource utilization.

Project Timeline and Phased Strategy (16 Weeks)

The **16-week project** is executed through **six distinct, sequential phases**, ensuring that each critical component is built upon a **stable foundation** laid in the preceding phase. This approach guarantees **systematic development, thorough quality checks, and minimizes technical debt**.

Project Timeline Breakdown

Phase	Title	Weeks	Key Activities
1	Planning & Design	W1 - W2	Requirements Gathering, System Design, UI/UX Wireframing.
2	Development - Foundation	W3 - W5	Backend & Frontend Setup, Authentication, User Management, Admin Panel foundation.
3	Core Features	W6 - W10	Teacher Management, Session Booking, Payment Integration, Real-time Chat System.
4	Advanced Features	W11 - W13	Dashboard implementation, Search/Filter, Newsletter, Python Interpreter, Image Management.

5	Testing & Refinement	W14 - W15	Unit, Integration, and User Acceptance Testing (UAT), Bug Fixes.
6	Deployment & Documentation	W16	Production Deployment, Documentation, Project Handover.

Phase Focus and Deliverables

Planning & Design (W1-W2): Establishes the system blueprint by finalizing user stories, creating the Database Schema, and producing high-fidelity UI/UX wireframes.

Development - Foundation (W3-W5): Erects the core architecture, setting up the MERN stack environment, implementing secure Authentication (JWT), and establishing user profiles and the administrative backend.

Core Features (W6-W10): Delivers primary value, focusing on Teacher Management (CRUD), complex Session Booking logic, secure Payment Integration, and real-time Chat System infrastructure (WebSockets).

Advanced Features (W11-W13): Enhances utility by building interactive Dashboards, dynamic Search/Filter algorithms, integrating the specialized Python Interpreter for coding practice, and managing image uploads (Cloudinary).

Testing & Refinement (W14-W15): Focuses on quality assurance through Unit testing, Integration testing, and crucial User Acceptance Testing (UAT) to confirm original business requirements.

Deployment & Documentation (W16): Concludes the project by moving the stable codebase to the production environment, finalizing comprehensive technical documentation, and preparing for user training and handover.

Resource Allocation (Person-Weeks)

The project leverages a **Two-member, cross-functional team**. The resource model is designed for efficiency, utilizing a high concentration of full-time developers during core build phases and targeted part-time assistance for specialized functions (Design, QA).

Manpower Structure and Detailed Responsibilities

Role	Count	Duration	Key Responsibility
Project Manager	1	Full-time	Coordination, Scope Control, Risk Mitigation, Client Communication.
Backend Developer	1	Full-time	Designing RESTful APIs, securing endpoints, database optimization, server-side logic.
Frontend Developer	1	Full-time	Developing React components, state management, ensuring responsive UI/UX across devices.
Full-Stack Developer	1	Full-time	Bridging front-end and back-end integration points, utility development, rapid prototyping.
UI/UX Designer	1	Full-time	Creating final design assets, running usability tests, ensuring visual consistency.
QA Tester	1	Full-time	Test case creation, execution, bug tracking, and verifying functional requirements.

Technology Resources and Strategic Cost Management

The technology stack selection is driven by **performance**, a **simplified development environment**, and **cost-effectiveness**, aligning with the stringent budget constraint.

Technology & Hosting Summary

Category	Resources	Notes for Budget Constraint
Stack	Node.js, Express, React, MongoDB	MERN Architecture: Provides a unified JavaScript environment, maximizing developer efficiency and reducing context switching time.
Tools	Git, VS Code, npm, Postman, Figma/Adobe XD	Utilizing community editions and open-source tools to minimize licensing costs.
Hosting & Services	Cloud Hosting (minimal shared-tier), MongoDB Atlas (free tier), Cloudinary (free tier), Domain & SSL.	Strategic reliance on permanently free tiers for MongoDB and Cloudinary is crucial for keeping infrastructure costs minimal and predictable in the early stage.

Project Schedule (GANTT Chart Analysis)

Task	Start Week	End Week
Planning & Design	W1	W2
Backend Setup	W3	W5
Frontend Setup	W3	W5
Authentication	W4	W6
User Management	W5	W7
Teacher Management	W6	W8
Session Booking	W7	W9
Chat System	W8	W10
Dashboards	W9	W11
Additional Features	W11	W13
Testing	W14	W15
Documentation	W15	W16
Deployment	W16	W16

Budget Estimation (In Bangladeshi Taka - BDT)

The project budget is strictly managed to a total estimated cost of **1,32,500 BDT**, ensuring the 1,50,000 BDT ceiling is respected. This strategic success stems from minimizing infrastructure overhead and structuring the team based on efficient, localized labor costs.

Cost Breakdown & Summary

Category	Item/Role	Cost (BDT)	Notes
Personnel (Total)	PM, Developers, Designer, QA	1,10,500	Based on local part-time/efficient rates.
Infrastructure (Total)	Hosting, Domain, Tools	22,000	Minimized costs using free tiers for DB/Storage.
Total Project Cost		1,32,500 BDT	The project is viable and under budget.